improper digestion and absorption, insufficient distribution of nutrients, accumulation of excess waste products lead to poor nourishment of cells.

With increasing age, cell functioning is slowed down and their response to hormones and synthesis of enzymes decreases. These changes are associated with a decrease in the number of functioning cells. Since the cells are interdependent, if one cell dies, others of the same or different organ lose their ability to function resulting in malfunctioning of the organ. This in turn gradually affects the total body functioning leading to death.

NUTRITIONAL NEEDS DURING OLD AGE

Energy

The calorie intake should be adjusted to maintain weight. The energy requirement decreases due to the following reasons:

a. Decreased physical activity
b. Reduction in lean body mass and increase in adipose tissue
c. Decrease in resting metabolic rate by 15-20 percent due to changes in body composition and physical inactivity

Protein

The protein requirements do not change during old age. It is the same as adults i.e. 1g/kg of body weight.

Carbohydrate

Since the calorie requirement is reduced, the carbohydrate content should also be proportionately reduced. Due to impaired glucose tolerance and gastrointestinal disturbances like constipation, emphasis should be on taking complex carbohydrates.

Fat

The fat requirements are also reduced, corresponding to the energy requirements. The intake of saturated fats and cholesterol should be less and unsaturated fat should be used in cooking.

Minerals

Calcium needs increases during old age due to increasing mobilization of calcium from bones and incidence of osteoporosis. During old age 1000mg of calcium is recommended per day because of the following reasons.

- Limited availability of calcium from foods
- To compensate age related bone loss and to improve calcium balance
- To prevent fractures and tooth decay
- To compensate decreased efficiency of calcium absorption

Aging does not affect iron needs. Hence iron needs are same as that for adults. Mild anaemia may affect the health of old people due to inefficient circulation. Therefore iron intake should be adequate during old age.

Vitamins

Vitamin A requirements remain the same i.e., 600 µg of retinol. Inadequate exposure to sunshine may affect vitamin D levels. The antioxidant vitamins such as vitamin E, carotenoids and vitamin C have been identified to promote health of the elderly. Vitamin B6 requirements are also increased due to gastritis which interferes with absorption.

Besides these various nutrients, water should be consumed in plenty as such or as fluids like buttermilk, fruit juice and soups. Intake of sufficient fluids reduces

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